Can Targeting Nudges Improve enrollment in California's ACA Health Insurance Marketplace? A Machine Learning Approach

By Roa'a Hamed*

This paper investigates whether targeting informational interventions ("nudges") to households, based on their observed characteristics, increases enrollment in California's Affordable Care Act (ACA) marketplace – Covered California – relative to traditional, one-size-fits-all approaches. I apply policy learning, a causal machine learning method, to estimate the optimal policy rule. I find moderate treatment effect heterogeneity, but targeting does not outperform simply sending every household the single best-performing letter. These results suggest that targeting nudges offers little advantage over existing approaches, and that closing enrollment gaps likely requires different policy tools altogether.

Nearly 60 percent of people eligible for affordable health insurance coverage under the ACA do not enroll (Kaiser Family Foundation, 2026). This persistent gap reflects, in part, behavioural frictions – search and psychological costs – that deter participation. Nudges like deadline reminders or plan leaflets can mitigate these frictions. However, because the ability to process information varies widely across people, there is growing interest in policy learning (or targeting) to match people to the interventions that work best for them.

Despite growing interest in this kind of targeted, data-driven allocation of policies in other domains like labour market programs (Frölich, 2008; Lechner and Smith, 2007), vaccination campaigns (Kitagawa and Wang, 2023), and naturalization programs (Ahrens et al., 2024), there is scarce evidence in the health insurance context. Which nudges work best, and for whom? Would targeting be more effective than a one-size-fits-all approach? While studies show that responses to nudges vary by household characteristics, it is unclear if this variation can be leveraged for targeting.

This paper investigates whether targeting nudges to households increases health insurance enrollment relative to one-size-fits-all approaches. My context is Covered California's 2016 open-enrollment period. I apply multi-action policy learning (Zhou, Athey and Wager, 2018) to experimental data (Domurat, Menashe and Yin, 2021*b*) to estimate an optimal policy rule that assigns households to a type of nudge based on their observed characteristics.

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Since its inception in 2010, the ACA has aimed to expand access to affordable and competitive health insurance and simplify consumer choice. However, consumers face behavioural frictions like a lack of awareness, search costs, inattention, and choice complexity that may impede their ability to gain coverage. Domurat, Menashe and Yin (2021*b*) designed four letters that provide increasing amounts of information to address these frictions. Relative to control (no letter), the letters increased enrollment by 1.0 to 1.5 percentage points (pp), with significant heterogeneous treatment effects by income: high-income households were less likely to enroll in response to more informative letters. These results suggest that sending every household the same letter may be a suboptimal approach for addressing low enrollment.

To evaluate whether targeting can improve enrollment, I apply the policy learning framework from Zhou, Athey and Wager (2018). I first use causal forests, a non-parametric method of estimating treatment effect heterogeneity (Athey, Tibshirani and Wager, 2019), to learn how each household's predicted enrollment varies with their observable characteristics. I then use these estimates to construct policy rules in the form of policy trees of various depths that assign one of the five letters (or no letter) to each household based on their characteristics, in order to maximize enrollment. Finally, I evaluate the performance of the derived policy rules against a set of alternative approaches such as "one-size-fits-all" policy of assigning the same treatment to everyone, and a model-free plug-in rule that assigns the treatment with the largest estimated treatment effect.

I find that, despite some evidence of heterogeneity, targeting is not more effective than a one-size-fits-all approach. The causal forest found moderate heterogeneity and potential for efficiency improvements: the distribution of predicted household treatment effects ranges widely from -8 to 14 pp, with the largest variation explained by household income, subsidy amount, age, and funnel round (i.e., the referral channel). However, the estimated policy rules and the plug-in rules led to a smaller, albeit statistically insignificant, gain in enrollment over the simple policy of sending every household the letter with the highest average treatment effect (Subsidy and Penalty). For example, assigning households nudges according to a depth-3 policy tree would lead to 0.434 pp less enrollment relative to sending everyone the best letter. Together, these results suggest that there is insufficient heterogeneity in response to the nudges to justify a targeting approach.

This study contributes to three strands of literature. First, it contributes to burgeoning research on the use of machine learning to estimate heterogeneous treatment effects (Athey, Tibshirani and Wager, 2019; Davis and Heller, 2017; Knittel and Stolper, 2021). This literature emphasizes the potential of using heterogeneity to improve targeting, however, few studies actually estimate actionable targeting rules. Second, I contribute to a growing application of policy learning to social programs (Cockx, Lechner and Bollens, 2023; Ahrens et al., 2024; Ida et al., 2026; Banerjee et al., 2025; Shah and Kreif, 2025) by providing the first

application to nudges in a health insurance context. Third, I contribute to the discussion on take-up of government programs by eligible people. While the literature is interested in the ability of nudges to impact vulnerable populations, evidence points to their limited effect. For example, nudges only have a limited ability to address them, such as non-English speakers (Myerson et al., 2022) and lower-income groups (Finkelstein and Notowidigdo, 2019; Bhargava and Manoli, 2015). Similarly, I contribute evidence that nudges fail to differentially increase enrollment by language, race, and gender. Policymakers interested in policy levers that reduce inequality in health coverage likely require alternative policy levers, such as auto-enrollment or in-person assistance, rather than nudges.

This paper proceeds as follows. Section I provides background information on the ACA marketplace and behavioural frictions therein, as well as the original experiment and treatments that I use for my analysis. Section II describes the data and its limitations. Section III details the research design, including the estimation of treatment effect heterogeneity and multi-action policy learning. Section IV presents the results, and Section V concludes.

I. Background

A. Covered California and Behavioural Frictions

Covered California, the state's ACA marketplace, was established to provide regulated and affordable health insurance to individuals without coverage through other means. To simplify consumer decision-making and improve affordability, plan characteristics and premiums are highly regulated; insurers in California can only vary plan premiums by age, and the ratio of premiums for the oldest to youngest consumers cannot exceed 3:1. Because of this community rating, age is a strong predictor of a household's premium, and information that reveals subsidy or price details may therefore be more consequential for older households facing higher sticker prices. Plans are also "guaranteed" issue, meaning no eligible consumer can be denied coverage.

Enrollment in ACA health insurance may be low due to the presence of large behavioural frictions. Surveys find that many uninsured households do not understand their own subsidy eligibility, and some remain unaware they qualify for coverage at all, particularly those transitioning out of Medicaid (Greenberg, 2017). Furthermore, enrollment is not automatic, so a household must apply, receive an eligibility determination, and then separately select and pay for a plan before the open-enrollment deadline. These issues reflect underlying behavioural frictions that can impede enrollment in the ACA marketplace.

B. The Covered California Experiment

Letter Interventions

To address these behavioural frictions, Domurat, Menashe and Yin (2021*b*) designed four nudges in the form of letters that progressively increased the amount of information. Households in the intervention arms received one of the following letters or Control (Arm 1; no letter):

- **Basic Letter** (Arm 2): stated the open-enrollment deadline, described the general benefits of insurance, and provided Covered California's website and phone number.
- **Subsidy and Penalty** (Arm 3): included the Basic Letter's content and added the household's personalized estimated monthly subsidy and tax penalty, calculated from the household's reported income and size.
- **Price Compare** (Arm 4): included Arm 3's content and added a table listing the Silver and Bronze plans available in the household's market along with their net-of-subsidy premiums.
- **Price-Quality Compare** (Arm 5): included Arm 4's content and added each plan's quality rating under the ACA's five-star Quality Rating System (QRS).

Each letter targeted a specific friction: the Basic Letter (Arm 2) addressed inattention to the deadline and low awareness of Covered California itself; the Subsidy and Penalty letter (Arm 3) addressed uncertainty about the household's own subsidy and tax liability; the Price Compare letter (Arm 4) addressed a lack of information on available plans and prices; and the Price-Quality Compare letter (Arm 5) added information on plan quality. Notably, Domurat, Menashe and Yin (2021*b*) did not directly test whether each letter reduced the friction it targeted, so I carry over this assumption to this study.

These frictions plausibly bind differently across households. Income and subsidy amount jointly determine a household's tax penalty exposure and net premium, so the Subsidy and Penalty and Price Compare letters should matter most for households with the least certainty about these figures, especially older households with higher premiums. Since all letters were sent in English or Spanish only, preferred language may limit their reach for other households. Furthermore, gender and race may matter given a broader literature that documents differential take-up of social programs by these characteristics (Myerson et al., 2022; Finkelstein and Notowidigdo, 2019; Bhargava and Manoli, 2015).

Study Sample

The sampling population ("Funnel") consisted of consumers that entered into Covered California from December 2015 to January 2016 during the open-enrollment

period. They were eligible for Covered California coverage in 2016 but had not selected a plan. Consumers entered the Funnel through two pathways: open-enrollment and county referrals.

These two populations may have different informational needs given their baseline awareness of Covered California before the interventions. Open enrollment applicants self-selected by applying to Covered California, but did not choose a plan before the study period for many possible reasons (e.g., behavioural frictions, other coverage). County referrals are applicants that lost Medicaid coverage due to a change of eligibility; their county informed them of their eligibility for a Covered California plan through a formal letter. As such, they may have had less awareness of the program compared to open-enrollment applicants.

II. Data

In this study, I analyze the public-use data from the Covered California experiment.[1] This includes administrative data on applications and enrollment from Covered California. The experiment randomly mailed one of five treatment arms (described above) to households and evaluated their effects on enrollment and adverse selection in the ACA marketplace.

I use the same post-exclusion sample as in the original experiment, drawn from the Funnel population. The final study population consists of 87,394 households.

I define the outcome, Y_i, as a binary indicator for whether household i enrolled in a Covered California plan during the study period.

I define the set of treatment options that the policy rule can assign to a household as $\mathcal{A} = \{0, 1, ..., D\}$, where $D = 4$, 0 denotes Control (no letter), and 1 through 4 correspond to the Basic Letter, Subsidy and Penalty, Price Compare, Price-Quality Compare letters, respectively. W_i is the household's actual assigned letter. Because assignment was randomized with equal probability across the five arms, the treatment propensities are known and equal across arms (0.2).

I observe a broad set of baseline household characteristics in the data, including household age (mean), income (% of FPL), maximum monthly subsidy amount (APTC), size, and number of kids. There are also indicators for e-mail availability, marital status, gender of the household head, race, preferred written language, California Rating Region (19 regions), and a Funnel round indicator. Across all covariates, I do not observe imbalances between treatment and control arms (see Table 1 for details).

Roughly 10 percent of households have missing income data. I impute them as zero, which is outside of the variable's distribution, and include a missingness flag for the policy tree analysis since the *R* `policytree` package does not accept missing values. I do not impute income when using the `grf` package since it has built-in capabilities to handle missing values.

A key limitation of the data is that I do not have access to the original study's

[1]The data repository is openicpsr-125801 (Domurat, Menashe and Yin, 2021*b*,*a*)

Table 1—Summary Statistics, by arm

	Treatment Arm						
	Control (1)	Basic Letter (2)	Subsidy + Penalty (3)	Price Compare (4)	Price-Quality Compare (5)	Total (6)	p-value all arms (7)
Gender	1.56	1.56	1.55	1.56	1.56	1.56	0.67
Married	0.60	0.60	0.60	0.60	0.60	0.60	0.84
Household Size	1.40	1.39	1.40	1.39	1.39	1.39	0.96
Number of Kids	0.11	0.10	0.10	0.10	0.10	0.10	0.74
E-mail Available	0.55	0.54	0.55	0.55	0.55	0.55	0.96
Income (% FPL)	212.40	211.98	212.45	211.89	212.52	212.25	0.86
Subsidy ($/month)	217.75	220.31	215.08	219.00	216.39	217.70	0.41
Missing Income	0.10	0.10	0.10	0.10	0.10	0.10	0.94
Latino	0.43	0.43	0.43	0.42	0.43	0.43	0.62
Black	0.05	0.05	0.04	0.05	0.05	0.05	0.31
White	0.26	0.26	0.26	0.26	0.25	0.26	0.62
Asian	0.12	0.12	0.12	0.12	0.12	0.12	0.79
Other Race	0.15	0.14	0.14	0.14	0.15	0.15	0.84
English	0.75	0.76	0.75	0.76	0.75	0.76	0.98
Spanish	0.17	0.17	0.17	0.17	0.17	0.17	0.89
Other Language	0.08	0.07	0.08	0.08	0.08	0.08	0.50
Observations	17,378	17,431	17,521	17,509	17,555	87,394	

Notes: Table 1 reports income and demographic characteristics reported in the administrative data for the final study sample, by treatment arm. Column 7 reports the p-value on the null that the mean across all arms are equal. p-values are computed from heteroskedasticity-robust (HC1) standard errors.

data on household health spending risk. Domurat, Menashe and Yin (2021*b*) use this data to conduct welfare analysis to understand where the letter interventions have a significant impact on the enrollment of high versus low risk consumers. Due to the absence of this data in the public-use replication file, I am unable to conduct a similar welfare analysis.

III. Research Design

A policy $\pi : \mathcal{X} \to \mathcal{A}$ maps household i's observed characteristics $X_i \in \mathcal{X}$ to an action from the set of possible actions $\mathcal{A}$. Policy learning aims to find the optimal policy rule π^*, within a candidate class Π, that maximizes expected enrollment in Covered California.

The section builds toward an estimate of π^* in three steps. I first estimate how treatment effects vary with X_i. Second, I use those estimates to construct household-level doubly-robust reward estimates and search over policy trees to estimate the optimal policy rule π^*. Third, I evaluate the resulting policy rule against alternative approaches.

A. Estimating Treatment Effect Heterogeneity

Applying Rubin (1974)'s potential outcomes framework, the conditional average treatment effect of letter a relative to Control is

$$\tau_a(x) = E\big[Y_i(a) - Y_i(0) \mid X_i = x\big], \qquad a \in \{1, \ldots, D\} \tag{1}$$

where $Y_i(a)$ is the potential enrollment outcome household i would realize under action a. We never observe this, but under random assignment, $\tau_a(x)$ can be estimated by comparing enrollment across similar households assigned to letter a versus Control.

I estimate $\tau_a(x)$ for all households and all letters using causal forests, a non-parametric machine-learning method for estimating heterogeneous treatment effects (Athey, Tibshirani and Wager, 2019). Causal forests split the covariate space repeatedly to maximize the difference, across splits, in the relationship between the outcome and treatment, uncovering how treatment effects vary across the sample. I use the `multi_arm_causal_forest` function from the `grf` package, with 3,000 trees and the default minimum node size of 5, and employ honest and out-of-bag estimation throughout. Honesty means the subsample of households used to decide where a tree splits is distinct from the subsample used to estimate the treatment effect within each resulting leaf, which prevents the forest from fitting splits to the same data used to report effects. Out-of-bag prediction means each household's predicted CATE is formed only from trees that did not use that household to grow their splits.

I fit the forest on the full vector of baseline household characteristics X_i available in the data, including income (% FPL), household size, number of kids, California Rating Region, household age (mean), gender of the household head, race, subsidy amount, e-mail availability, language preference, marital status, and flags for missing income and income above 1,000 % FPL.

I use these CATE estimates to describe heterogeneous treatment effects of the letters in three ways. First, I plot the distribution of CATEs for each letter. Second, I report the causal forest's variable importance measure for each letter, which is a measure of how often a covariate is split on.[2] Third, for the top covariates, I plot the CATEs to show how they compare to the average treatment effects (ATEs) of the letters.

B. Learning the Optimal Policy

Next, I examine whether any heterogeneity we found above can be exploited to target letters. This analysis proceeds in two steps. First, I construct a household-level, doubly-robust estimate of the enrollment each household would realize under

[2]The multi-armed forest pools importance scores across treatment arms, so I compute this separately for each letter by fitting four independent binary causal forests, each restricted to the Control arm and one treatment arm, with the within-subsample propensity of 0.5.

every action. Second, I use that estimate to search over a class of policy trees that maximize expected enrollment.

Estimating Cross-Fitted Doubly Robust Scores

Evaluating any candidate policy requires an estimate of the enrollment each household would have realized under every action, not just the one it actually received. I construct these using the Cross-fitted Augmented Inverse Propensity Weighted (CAIPW) scores (or doubly-robust scores) following Zhou, Athey and Wager (2018),

$$\hat{\Gamma}_{ia} \;=\; \hat{\mu}_a^{-k(i)}(X_i) \;+\; \frac{\mathbb{1}\{W_i = a\}}{e_a(X_i)}\Big(Y_i - \hat{\mu}_a^{-k(i)}(X_i)\Big), \qquad a \in \mathcal{A}, \tag{2}$$

where $\hat{\mu}_a$ is household i's expected enrollment under action a, W_i denotes the treatment arm actually assigned in the experiment, and $-k(i)$ indicates that the nuisance estimate is fit without using the fold containing observation i. The first term is the model-based prediction; the second corrects it using i's actual outcome. Since propensity scores are known, $\hat{\Gamma}_{ia}$ is unbiased even if $\hat{\mu}_a$ is misspecified (Robins, Rotnitzky and Zhao, 1994).[3]

I estimate $\hat{\mu}_a$ with a multi-arm causal forest fit on the full covariate set described above, with the default settings (3,000 trees, minimum node size 5), and implement K-fold cross-fitting following Zhou, Athey and Wager (2018): I randomly split the sample into 5 roughly equal folds and, following Athey, Keleher and Spiess (2025) Section C.2, supply the fold identifier to `grf`'s `multi_arm_causal_forest` function as a clustering variable, so the algorithm resamples whole folds and each household's out-of-bag prediction comes only from trees that excluded its entire fold. I extract $\hat{\Gamma}_{ia}$ from the fitted forest using `grf`'s `double_robust_scores` function, which implements (2) directly.

Fitting Policy Trees

I use the $\hat{\Gamma}$ from above to search, within a restricted class of policies Π, to estimate the optimal policy rule that maximizes expected enrollment in Covered California,

$$\hat{\pi} = \arg\max_{\pi \in \Pi} \left[\frac{1}{n} \sum_{i=1}^{n} \hat{\Gamma}_{i,\pi(X_i)} \right] \tag{3}$$

I employ the cross-validation process from above to evaluate the policy trees. For the households in fold k, I fit a policy tree on the $\hat{\Gamma}$ rows outside of it, and evaluate the policy rule $\pi(X_i)$ on the households in fold k. This ensures that a

[3] I use the augmented rather than the plain inverse-propensity estimator because the outcome model absorbs variance that the reweighting alone would leave in place.

policy tree never chooses a treatment for a household that it used to come up with the rule.

I choose Π to be the class of shallow decision trees, or policy trees. Policy trees have two advantages over black-box plug-in rules (introduced below), which assign each household the arm with the largest estimated effect and offer no account of what drives the assignment: they let a policymaker choose/exclude the variables used to in their policy rule (e.g., excluding race or gender), and they are easy to visualize and communicate to stakeholders.

A depth-L policy tree splits the covariate space through at most L nested splits and assigns a single treatment (or Control) to all households in each terminal leaf. I fit policy trees of depth 2 and 3 using an exact search (the `policy_tree` function of the *R* `policytree` package), and a depth-4 tree using `hybrid_policy_tree`.[4] To keep the exact search over depth-2 and depth-3 trees tractable, I discretize the continuous covariates (income, subsidy, age) into 20 quantile bins each, and search over every bin threshold (`split.step` = 1).

I fit the trees on a restricted set of household characteristics: mean household age, income (% FPL), the missing-income indicator, an indicator for income above 1,000 % FPL, the maximum monthly subsidy amount, e-mail availability, funnel round, household size, English language preference, and indicators for three California Rating Regions (Regions 3, 15, and 18). I select this set from the union of the per-arm top-ten covariates by variable importance (Section III.A). I deliberately exclude sensitive characteristics (gender, race).[5]

Policy Evaluation

I benchmark the estimated policy trees against three alternative approaches a policymaker might consider instead: 1) sending no letters; 2) one-size-fits-all approaches where the same letter is sent to everyone; and 3) the plug-in rule, which assigns households the letter with the largest predicted CATE – a black-box approach that personalizes treatment assignment more flexibly than a shallow policy tree.[6]

I report all comparisons as paired differences in estimated value (i.e., the gain in enrollment from a candidate policy). For any two policies, π and π', the estimate

[4] A hybrid search performs an exact search to a specified depth followed by a greedy search over any remaining levels. An exact search over all four levels of a depth-4 tree is combinatorially infeasible at this sample size. More information can be found in the package's *R* documentation (Sverdrup et al., 2026).

[5] The causal forest that produces $\hat{\Gamma}$ retains the full covariate set to retain the accuracy of the estimates, while allowing the policymaker to maintain transparency and privacy by excluding sensitive variables from the deployable policy trees.

[6] I estimate the plug-in rule with a separate multi-arm causal forest following the procedure in Section III.B.1, but excluding gender and race so that the rule remains fairness-constrained in the same way as the policy trees. I set a minimum node size of 300 to reduce noise in the estimates following Athey, Keleher and Spiess (2025).

enrollment gain is therefore

$$\hat{\Delta}(\pi, \pi') = \frac{1}{n} \sum_{i=1}^{n} \left(\hat{\Gamma}_{i,\pi(X_i)} - \hat{\Gamma}_{i,\pi'(X_i)} \right), \tag{4}$$

with standard errors computed from the standard deviation of the household-level differences in enrollment under each policy.

IV. Results

A. Heterogeneous Treatment Effects

In this section, I describe the treatment effect heterogeneity that I estimate using causal forests.

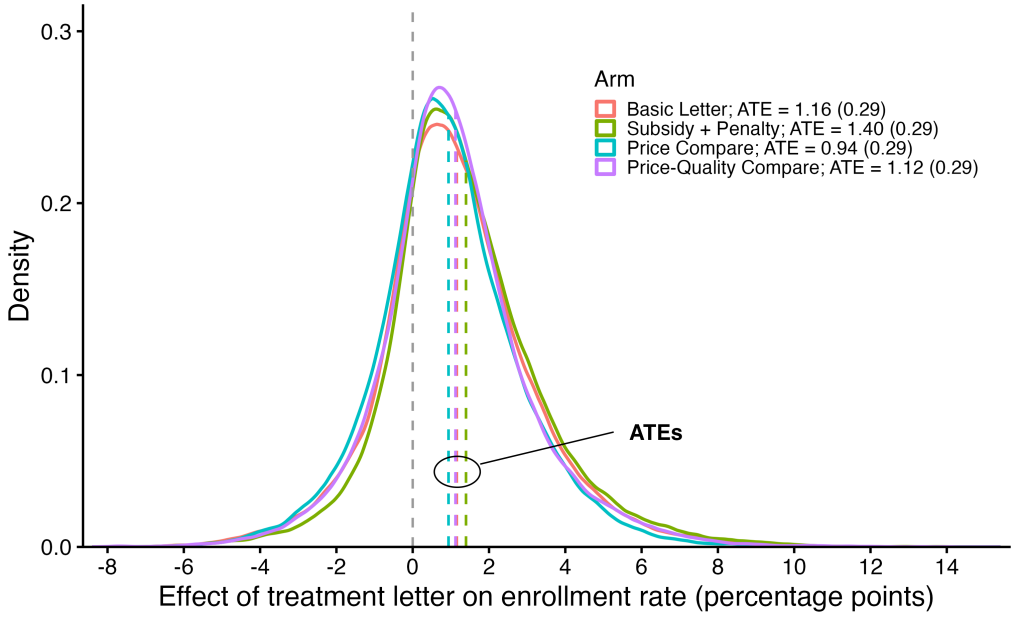


FIGURE 1. DISTRIBUTION OF HOUSEHOLD-LEVEL PREDICTED TREATMENT EFFECTS OF EACH LETTER, USING MULTI-ARM CAUSAL FORESTS.

Note: The horizontal axis measures the predicted change in enrollment during Covered California's open-enrollment period, in percentage points. Kernel densities of out-of-bag predicted conditional average treatment effects $\hat{\tau}(X_i)$ across all households, from a multi-arm causal forest. Dashed lines mark each arm's estimated average treatment effects (ATEs); standard errors in parentheses. Kernel smooth with Epanechnikov Kernel and Silverman (normality) bandwidth.

Figure 1 shows the distribution of estimated treatment effects for each letter.

First, I note that the causal forest is able to recover similar ATEs as the OLS specification in the original experiment. All letters have positive and significant treatment effects, ranging from 0.94 pp (SE = 0.29) for the Price Compare letter to 1.40 pp (SE = 0.29) for the Subsidy and Penalty letter.

Second, the histogram shows that treatment effects for each letter range from around −8 to 14 pp. These results provide suggestive evidence that while the letters had positive effects on enrollment on average, they may have dissuaded some households from enrolling. In that case, it may be justified to target the letters where their effects are positive in order to maximize enrollment.

Finally, these estimates are noisy, as evidenced by the large distribution of treatment effects away from the ATEs. However, estimates from the causal forests are theoretically honest, so we can still use them to construct and evaluate policy rules (Athey, Keleher and Spiess, 2025).

Variables explaining treatment effect heterogeneity

Next, I examine which household characteristics are driving treatment heterogeneity. Figure 2 shows the average and conditional average treatment effects for each letter for the four most important variables: income (% FPL), subsidy amount, household age (mean), and funnel round (see Appendix Figure A1, Panel A for more details).

I plot the CATEs for these top variables in Figure 2 to examine how treatment responsiveness varies across different household subgroups. The results suggest there is some treatment effect heterogeneity. Relative to other letters, the Basic Letter appears to be more effective for higher income (250–400%) and medium subsidy ($101–350) households. The Subsidy and Penalty letter may be more effective for lower income (150–200%), low subsidy ($1–100), young (<26) and open-enrollment households. The Price Compare and Price-Quality Compare letter only show some heterogeneity for low-income (<150%) and middle-aged households (26–40), respectively.

Notably, I find very little heterogeneity by sociodemographic variables that the literature associates with behavioural frictions, such as race and language (Appendix A2). The Basic Letter appears ineffective for heads of households that are Black households or prefer Spanish. By contrast, it appears most effective for the "Other" race group.

Finally, I note that the estimates sometimes have wide confidence intervals that overlap across letters. As such, these results only provide suggestive evidence of treatment effect heterogeneity. In the next section, I use these CATE estimates to estimate and evaluate policy rules; significant results will indicate whether there is sufficient treatment effect heterogeneity to justify targeting letters.

B. Multi-Action Policy Learning

In this section, I present the results from learning the optimal policy and evaluate whether it outperforms alternative approaches.

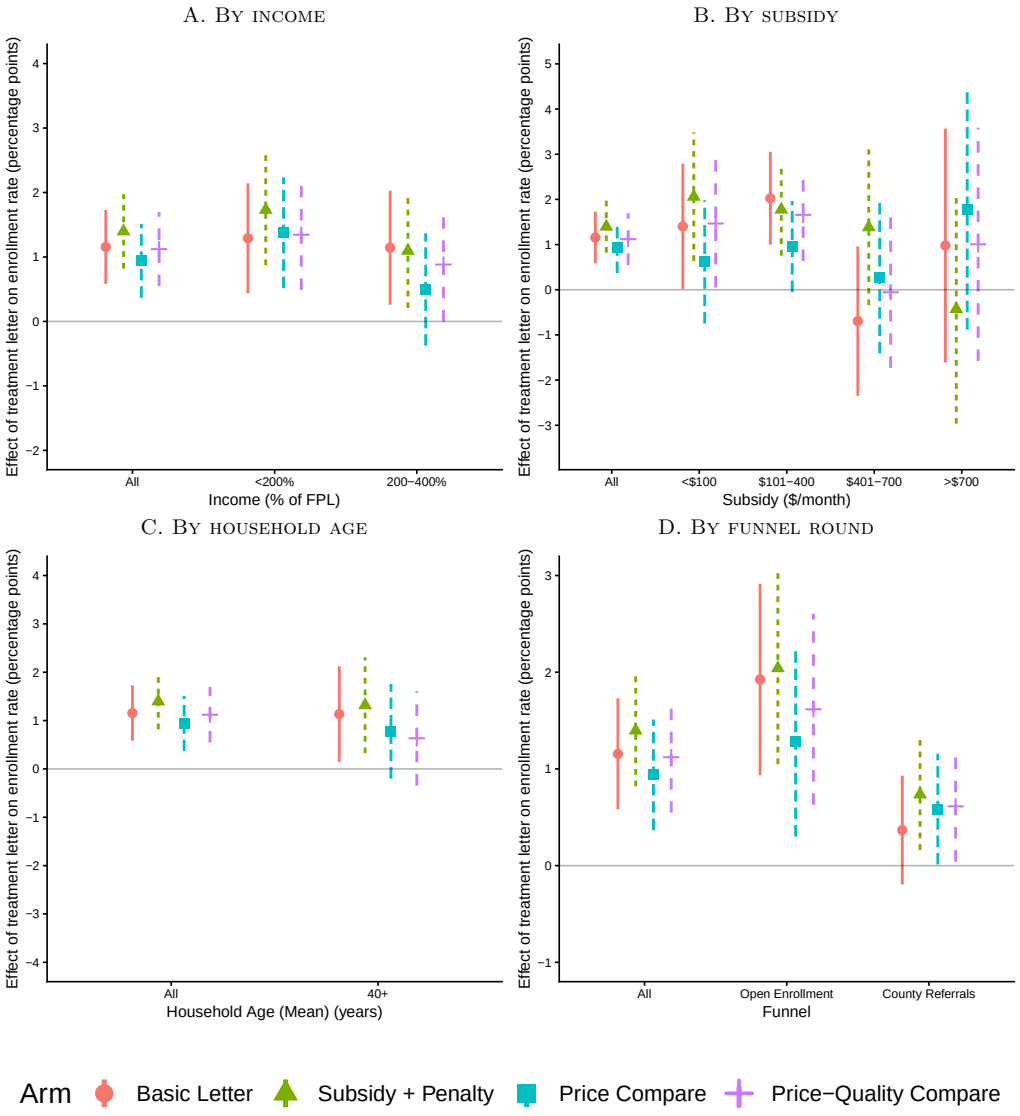


FIGURE 2. AVERAGE AND CONDITIONAL AVERAGE TREATMENT EFFECTS, USING MULTI-ARM CAUSAL FORESTS

Note: Figure 2 shows the average and conditional average treatment effects by household covariates. Displayed are the top four covariates explaining treatment heterogeneity in household enrollment in a Covered California health insurance plan: income (% FPL), subsidy, household age (mean), and Funnel round. Displayed estimates (with 95% confidence intervals) are honest and out-of-bag using *R* `multi_arm_causal_forest`.

Learning the Optimal Policy Rules

Figure 3 displays the optimal policy rule in the form of a shallow depth-2 tree.[7] This policy rule suggests that most of the treatment effect heterogeneity is driven by subsidy, income, and household age; for example, the policy rule assigns households with low subsidy amounts ($\leq$\$129) that are older ($\geq$37 years) the Subsidy and Penalty letter, but assigns younger ones (<37 years) the Price-Quality Compare letter. When I estimate a depth-3 policy tree, it splits on the same covariates (subsidy, age, income), as well as the Funnel indicator (see Appendix V). These results are consistent with the heterogeneity analysis in Section IV.A.

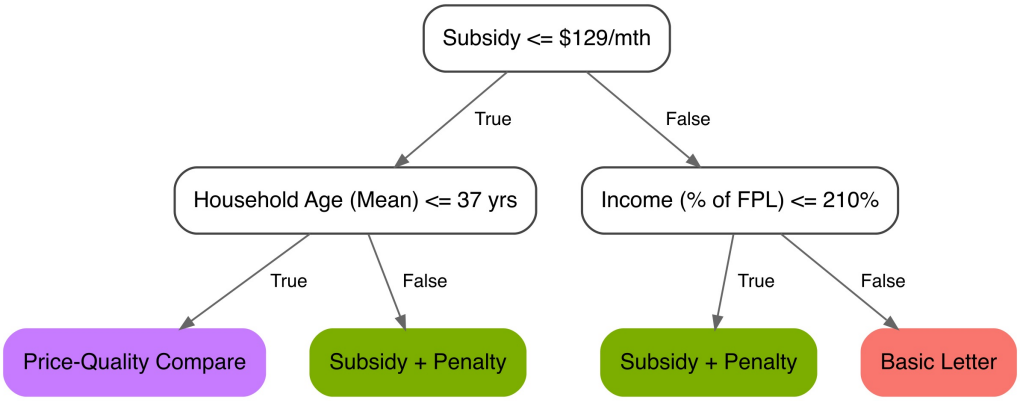


Figure 3. Fitted depth-2 policy tree.

Note: The figure shows the policy tree fitted to all 87,394 households in the data. The policy trees are built using the *R* `policytree` package and assign households in each terminal leaves the indicated policy in order to maximize expected enrollment of the sample in a Covered California health insurance plan.

Evaluating the Policy Rules

Finally, I evaluate the out-of-sample performance of the optimal policy rules and present the results in Table 2. In each column, the table reports the gain in reward $\hat{\Delta}$ of a specific policy choice compared to alternative policy rules (shown in rows). For example, the coefficient 1.131 (SE = 0.292) in the top-left cell is the gain in enrollment from assigning the Basic Letter to every household in the sample relative to the policy of assigning no letter.

The results show that none of the policy trees significantly outperform any one-size-fits-all policy, except sending everyone Control (no letter). In fact, all

[7]Note that the displayed policy trees are presented solely to illustrate the estimated decision rules. I fit them using the full sample of households. I also estimate a depth-4 policy tree (not displayed).

TABLE 2—ESTIMATED REWARD OF THE POLICY RULES COMPARED TO ALTERNATIVE APPROACHES.

	One-Size-Fits-All				Policy Tree			
	Basic Letter	Subsidy + Penalty	Price Compare	Price-Quality	$d = 2$	$d = 3$	$d = 4$	Plug-in Rule
Control	1.131 (0.292)	1.402 (0.294)	0.930 (0.292)	1.141 (0.293)	0.963 (0.291)	0.968 (0.288)	1.113 (0.288)	0.900 (0.290)
Basic Letter		0.271 (0.303)	−0.201 (0.301)	0.010 (0.302)	−0.168 (0.244)	−0.162 (0.261)	−0.017 (0.260)	−0.230 (0.268)
Subsidy + Penalty			−0.472 (0.303)	−0.261 (0.304)	−0.439 (0.227)	−0.434 (0.231)	−0.289 (0.244)	−0.502 (0.195)
Price Compare				0.211 (0.302)	0.033 (0.288)	0.039 (0.284)	0.184 (0.283)	−0.029 (0.293)
Price-Quality					−0.178 (0.281)	−0.172 (0.265)	−0.027 (0.260)	−0.240 (0.278)
Policy tree ($d = 2$)						0.006 (0.216)	0.151 (0.208)	−0.062 (0.206)
Policy tree ($d = 3$)							0.145 (0.197)	−0.068 (0.216)
Hybrid tree ($d = 4$)								−0.213 (0.218)

Notes: Table 2 reports the difference in estimated rewards between policy rules (columns) relative to alternative approaches (rows). The results are based on a 5-fold cross-validation exercise where I repeatedly fit the policy trees and plug-in rule on all but one fold, and evaluate it on the held-out fold. I report the average differences in rewards and associated standard errors (in parentheses). The One-Size-Fits-All policies assign every household to a single letter only; policy tree policies are depth-d trees using the R `policy_tree` function. I use `hybrid_policy_tree` to fit the hybrid policies, which is an approximate tree search. The policy trees are fit on a restricted set of household characteristics, including mean household age, income (% FPL), maximum monthly subsidy amount, e-mail availability, funnel round, household size, whether English is the head of household's preferred language, flags for missing income and income above 1,000 % FPL, and dummies for California Rating Regions 3, 15, and 18. The plug-in rule is fit on additional covariates: number of household kids, marital status, written language preference, and dummies for all 19 California Rating Regions.

three policy trees under-perform relative to always sending everyone the Basic Letter, Subsidy and Penalty, or Price-Quality Compare letters. However, the only significant differences are between the Subsidy and Penalty letters and the depth-2 and depth-3 policy trees (p-value < 0.10). For example, if policymakers deployed the depth-3 policy tree, it would lead to 0.434 pp (SE = 0.231) less enrollment relative to just sending everyone the Subsidy and Penalty letter.

We would expect that the plug-in rule, the most personalized policy, would outperform both one-size-fits-all policies and policy trees; however, this is not the case. The results show that the plug-in rule under-performs relative to every alternative policy approach except Control. However, the coefficients are small and insignificant.

Together, these results suggest that targeting letters is ineffective relative to one-size-fits-all policies. The policy with the largest gain we observe is in sending everyone the Subsidy and Penalty letter, which increases household enrollment

by 1.402 pp (SE = 0.294) more than Control.

Limitations of Non-Parametric Models

Why does policy targeting not outperform a one-size-fits-all policy?

There may be a fairly low signal-to-noise ratio in the study sample. In general, non-parametric models like policy trees and plug-in rules do not perform well when treatment effects do not vary much or when there are too many high-dimensional covariates (Athey, Keleher and Spiess, 2025). In these cases, it is difficult for causal forests to estimate the true CATEs well, resulting in policy rules fitting to noise in the data rather than exploiting true heterogeneity, if it exists. Several pieces of evidence point to this explanation. Figure 1 showed a large distribution of treatment effects from the causal forest, suggesting there was a lot of noise in the data. Furthermore, there was only moderate treatment heterogeneity for the top covariates in Figure 2, and many estimates had large, overlapping confidence intervals. These results suggest that there was limited heterogeneity in the sample for the policy rules to exploit.

Moreover, the average treatment effects of the letters are generally small, so any heterogeneity would have to be fairly large to be detected by the causal forest. Given these results, the household characteristics in the data do not seem to be strong predictors of treatment effect heterogeneity.

V. Conclusion

This study examined whether using machine-learning to target nudges can improve health insurance enrollment relative to one-size-fits-all approaches. While there is some heterogeneity in how households respond to different nudges, especially by age, income, subsidy, and Funnel round, the policy rules do not increase enrollment relative to the simple policy of sending everyone the most effective letter (Subsidy and Penalty).

Notably, I find little heterogeneity by language, race, or household size, characteristics the literature associates with inequality in enrollment. One interpretation is that these letters are not the right instrument for the frictions facing these groups. Since the letters address deadline awareness, subsidy uncertainty, and price information, a flat response by language or race may reflect a mismatch between the friction the letter targets and the friction that actually binds, rather than the absence of a friction altogether. This is consistent with prior evidence that nudges have limited reach for non-English speakers and lower-income households (Myerson et al., 2022; Finkelstein and Notowidigdo, 2019; Bhargava and Manoli, 2015), and suggests other policy approaches (e.g., auto-enrollment) are needed to close enrollment gaps.

Three limitations qualify these results. First, the sample is drawn from California's 2016 open-enrollment period; changes to the ACA since then, including the removal of the tax penalty, may limit how well these results generalize to

other states or periods. Second, the letters' average treatment effects are small and there is a fairly low signal-to-noise ratio in the study sample. Third, I lack health spending risk data to assess welfare, which goes beyond simply maximizing enrollment.

Future work could examine whether nudges can be effectively targeted in settings with larger samples or more heterogeneous populations. More broadly, policymakers and researchers alike may be better served by focusing on approaches that remove structural barriers to enrollment, rather than refining nudges.

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Appendix

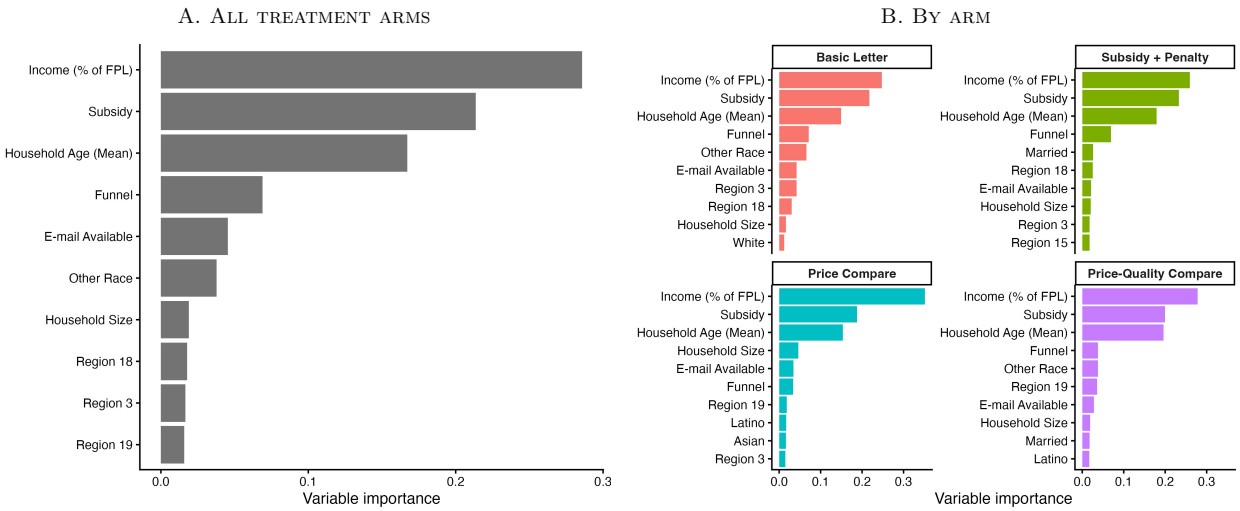


Figure A1. Heterogeneity in Treatment Effects: Variable Importance

Notes: The figure shows the top ten covariates that explain treatment heterogeneity in the enrollment of a Covered California health insurance plan, by importance. Panel A reports variable importance, using multi-arm causal forests fit on all 87,394 households in the sample. Panel B reports importance from separate causal forests estimated on the subsample restricted to Control and one treatment arm. Variable importance is a weighted measure of how frequently each covariate is split on, with splits nearer the root of the tree weighted more heavily. Since importance scores come from different forests estimated on different subsamples, only the relative ranking of covariates within each panel is meaningful. I use `grf`'s `variable_importance` function to estimate all importance measures on forests generated from its `multi_arm_causal_forest` (Panel A) and `causal_forest` (Panel B) functions. All causal forests are fit on the full set of household characteristics, including household size, number of kids, household age (mean), gender of the household head, race, language preference, marital status, subsidy amount, household income (% FPL), California Rating Region, funnel round, e-mail availability, and flags for missing income and income above 1,000 % FPL.

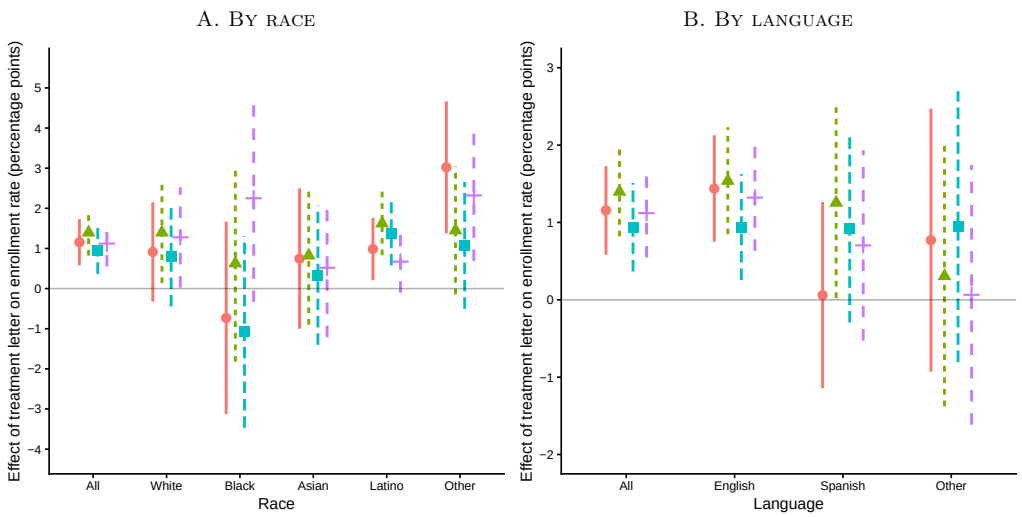


Figure A2. Average and conditional average treatment effects, using multi-arm causal forests

Note: The figure shows the average and conditional average treatment effects by household covariates. Displayed estimates (with 95% confidence intervals) are honest and out-of-bag using *R* `multi_arm_causal_forest`.

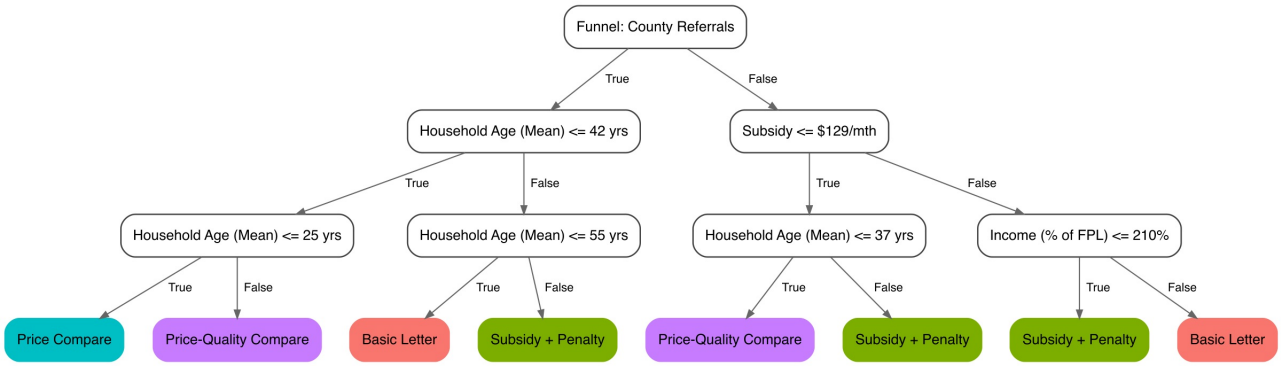


Figure A3. Fitted depth-3 policy tree.

Note: The figure shows a depth-3 policy tree fitted to all 87,394 households in the data. The trees are built using the R `policy_tree` function, and assign households in each terminal leaves the indicated policy in order to maximize expected enrollment of the sample in a Covered California health insurance plan.